Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Deep Learning, Custom Neural Architectures, Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Forward Deployed Machine Learning Engineer – Stratum AI

Dear Hiring Manager,

I am applying for the Forward Deployed Machine Learning Engineer role at Stratum AI. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia. I build deep learning systems end-to-end and deploy them where the data is messy and the client is on site. Models that tell a mine what is underground, and that stakeholders must trust, are the kind of problem I want to work on.

Three parts of your posting match my work. For custom neural architectures in PyTorch and Python, I design and train my own models, from stacked RNNs for sequential data to CNNs, and I benchmark them with baselines and ablations. My published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline, with F1 0.76 (Journal of the Audio Engineering Society, 2026). For applied work with clients, at MAS Holdings I led computer vision systems deployed in factories, including care-label inspection that cut inspection time by 99.5% and raised operational efficiency by up to 300%, and a vitals-monitoring system deployed in hospital wards. For communication, I have presented results to non-ML partners and stakeholders throughout, and I track deployed models with monitoring.

I should be direct about the gaps. My deep learning work has centered on CNNs and recurrent networks, and I have not built graph neural networks or large transformers from scratch; I have used HuggingFace models at a working level. I have not worked in mining or used FastAPI, DynamoDB or Azure in production, though I have run Python services and AWS pipelines. I expect to learn the domain and stack quickly, and I am comfortable with on-site client work.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience can support the Technical Services team at Stratum AI. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD